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COVER STORY

Sawing a Dam in Half (on Purpose)

[Note that this article is a transcript of the video embedded above.]

Concrete is the second-most-consumed substance on our planet.

Wesley Crump · Practical Engineering (blog)

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Concrete is the second-most-consumed substance on our planet. Only water beats it, and actually water is a major ingredient of concrete anyway. Every year, humanity mines, mixes, and places roughly three metric tons for every person on Earth. It's ubiquitous. Most of us hardly even think about all the concrete around us. We've all seen the grey lumpy mixture flowing down chutes into formwork to become a road, sidewalk, footing, pile, patio, or foundation. It's easy to think of concrete as a single, uniform substance used around the world. But it's not.

The only reason we are able to use so much concrete in construction is that it's cheap. Of the four main ingredients - sand, gravel, cement, and water - two of them come directly from the ground with little need for processing or refinement. One is water. Cement is the only ingredient that requires a significant manufacturing process, but the raw materials for it are fairly widespread across the globe. Many building materials are constrained by geography. They only grow, occur mineralogically, or are manufactured in specific locations. Then they have to be transported, often at great cost, to where they're needed. It's not true for concrete. No matter where you are on earth, there's a pretty decent chance that somewhere nearby exists a ready source for at least most of the raw ingredients you need to make it. That simple fact has significantly contributed to its widespread use, but it's done something else too.

Take a look at any geologic map. If you're like me, you do this in your spare time anyway. You realize pretty quickly that there is tremendous variability in the different kinds of materials that make up the surface of Earth's crust. And the practical result of that, at least for the purposes

of this discussion, is that every batch of concrete is just a little bit different depending on where you go. In a way, that's kind of special, right? In most cases, the concrete you see around you represents a particular place on Earth. Its strength, durability, appearance, and essence are highly local characteristics. It's literally made from materials that were sourced not too far away. But, in some cases, we've learned too late that local materials had some hidden problems when used in concrete, and the ways we've worked to fix those problems have created some of the most interesting stories. I'm Grady, and this is Practical Engineering.

This is Fontana Dam on the Little Tennessee River in North Carolina. At 150 meters (or nearly 500 feet) in height, it's the tallest dam east of the Mississippi River. The north shore of Fontana Reservoir forms the border of the Great Smoky Mountains National Park. And if you're through-hiking the Appalachian Trail, the famed 2200-mile path through the wildest parts of the eastern United States, you have to walk right over the top of it. Built by the Tennessee Valley Authority (or TVA), Fontana was completed in 1944, just in time to provide hydropower to the Alcoa aluminum smelting plant at the end of World War II. It's a concrete gravity dam, meaning that it derives its stability to hold back Fontana Reservoir entirely from its own weight. And boy does it have a lot of weight. More than 2.1 million cubic meters of concrete went into the structure before it was finished. That's well over half the volume of Hoover Dam, and if you watch the same kinds of videos I do, you know that putting all that concrete in Hoover Dam was a major challenge.

Concrete heats up as it hardens, which can negatively affect the curing process, but more importantly, it

causes the concrete to expand. For a structure like a dam sandwiched between two rocky abutments, that expansion can lead compressive stress to build up in the concrete. Then, after curing, when the concrete starts to cool back down, it shrinks. That shrinking can lead to cracks, especially in mass concrete structures that heat up and cool down unevenly. And cracks are not ideal for dams. To mitigate this issue, pipes were installed within the concrete at Hoover Dam, and chilled water was continuously circulated during construction to pull heat out of the concrete. The same thing was done when they were building Fontana Dam. In fact, in addition to the cooling lines, the dam was built with deliberate expansion joints that would allow each separate concrete block to cool off and shrink. Once the concrete cured, those joints were grouted to add strength and make the dam watertight. It was a pretty robust and thoughtful plan to avoid the buildup of stress in the structure, or so they thought.

In 1972, engineers inspecting the drainage gallery, a tunnel through the concrete dam used to collect and redirect drainage, noticed unexpected cracks right where the dam curves. Later investigation revealed that the cracks extended through a large part of the structure. At this point, the dam was still less than 30 years old. It shouldn't be deteriorating this quickly. But the cracks were serious enough that something needed to be done.

Engineers initially blamed the Tennessee sun. Fontana Dam runs almost perfectly east to west, with its broad downstream facing directly south. That means a huge area of concrete is exposed to sunlight for most of the day. The sun heats the concrete, causing it to expand, and over thousands of cycles, cracks are inevitable. The curved section of the dam was most vulnerable. Reaction forces from the abutments align with the axes of the dam. Instead of pure compressive stress, the expansion of the concrete created bending stress (a combination of expansion and contraction) at the corner. In addition to the cracks, the movement was also causing the spillway gates to bind up.

After instruments were installed on the dam, the scope of the problem became clear. Thermal movement is cyclical with the seasons. Concrete may expand in the summer, but it returns to its original size in the winter as temperatures cool. Fontana had some of that, but underneath the cyclical changes was a continuous one. The concrete was permanently growing.

TVA took some cores of the concrete to start planning a repair, and sent them out for testing. When the results came back, the reason for the unexpected growth was discovered. The laboratory that examined the concrete under the microscope noticed that some of the aggregates inside had dark rims around them. That is a classic sign of alkali-silica reaction, or ASR, sometimes known as concrete cancer.

The fundamental components of concrete are aggregates, large and small, bound together by a paste of

cement and water. As the cement paste hydrates, potassium and sodium hydroxides dissolve into the water within the tiny pore spaces of the concrete, creating an alkaline solution. In some cases, this is a good thing. The alkaline environment is great for steel reinforcement, helping to prevent rust. But for some types of aggregates, it causes a serious problem. Specifically, if reactive forms of silica are present, they can more readily dissolve in the high-pH water, combining with the alkalis to form a kind of gel. As that gel absorbs moisture, it swells and expands, causing internal stress and cracking.

This is an extremely widespread problem that has caused structural damage in every state in the US and many countries around the world. You usually don't have to search far for an example of a cracked up bridge, broken sidewalk, or ruined building foundation that resulted from an alkali-silica reaction in the concrete. Fortunately, the reaction requires three conditions, so there are quite a few ways to deal with it.

For one, an alkali-silica reaction requires the aggregates to actually contain silica, also known as silicon dioxide. Well, 90 percent of the Earth's crust is made up of silicate minerals, so this might not seem possible to avoid. Luckily, only certain forms of silica are significantly reactive in concrete. We have tests we can perform ahead of time to identify quarries or sources of rock that react with cement, allowing us to just avoid the issue altogether. But like I mentioned before, the cost of concrete is really sensitive to transportation costs. The farther you have to go to get suitable aggregates, the higher the project's costs rise, so avoiding local materials is not always ideal.

The second condition required for an alkali-silica reaction is highly alkaline cement. So, we have ways to control for that too. Cement can be manufactured to have lower alkali content, and we can use what are called "Supplementary Cementitious Materials," like fly ash, to replace some of the cement in concrete. Those solutions only work if the concrete isn't already in place, though.

The third factor of an alkali-silica reaction is excess moisture. You can just keep the concrete dry with waterproof coatings or membranes. Without moisture, the gel can't expand, so the problem is solved. But there are some structures where waterproofing is a pretty big challenge. So TVA was in a bind, literally. They were facing the possibility of just having to perpetually repair cracks and equipment as Fontana continued to expand. Then they decided to get creative. Kristen Smith is the Senior Program Manager for Dam Safety at TVA, and she explained the thought process:

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Wesley Crump

class="">Kristen: “You know, the impacts on the spillway and powerhouse equipment. That led to major maintenance and repairs [...] Need to move from the reactive approach - that’s not a long-term solution - to a more proactive approach.”

class="">The proactive approach they landed on was a fourth option for dealing with ASR: Rather than trying to stop the reaction, TVA decided to just give the concrete more room to grow. The solid rock abutments at each end of the dam had no room to give, so that space would have to be found in the dam itself.

class="">In 1976, they embarked on a fairly novel operation to cut a relief slot all the way through Fontana Dam and do it without draining the reservoir or causing any disruptions to the hydropower plant. The idea was pretty simple: instead of building up axial stress as the concrete expands, the dam can expand into the newly cut slots. Simple in theory; pretty challenging in practice. How do you saw a dam in half? Luckily, TVA has done this at two of its other dams in addition to Fontana, and shared some footage of that so you could see it happen.

class="">These are big dams, so this isn’t sawing with blades you find at a hardware store. The tool used for cutting through the concrete looks more like a rope than a saw blade.

class="">Kristen: “It is diamond wire, and it’s really neat. It’s, if you touch it, you know, it’s 15 millimeters, which is a little over half an inch. It’s abrasive. I mean, you know, it would rub your skin if you drag it across your skin, but you can touch it. You can run your hand along it and it’s not going to cut you. It can cut through concrete. It can cut through steel. [...] It looks like a big necklace.”

class="">That big diamond necklace runs along pulleys strategically installed on the dam to advance the slot downward. The saw pulls the wire in a loop, managing the slack and keeping constant tension against the bottom of the slot. There are a lot of advantages to this, in addition to the practically unlimited depth. It causes very little vibration or dust, and provides a clean cut without breaking the edges. But, there’s a pretty obvious challenge of cutting a slot in a dam: how do you deal with the water?

class="">Turns out, it depends on the dam. At Fontana, crews installed a cofferdam on the upstream face of the dam to hold back the reservoir during the operations. It’s basically half of a steel pipe that seals against the concrete face on the sides and bottom, just big enough for access

to adjust the pulleys. At Chickamauga Dam, the geometry made a cofferdam less feasible. So instead, they broke the process up into three sections separated by boreholes drilled downward into the structure. One section could be cut by the diamond wire while the other borehole was sealed, preventing water from moving through the slot. That’s easier said than done, but you can look to your feet for inspiration. The seals installed in the boreholes are long rubber tubes called sock seals.

class="">Kristen: “Well, it’s like a sock you put on your foot, but a half-inch thick rubber and a hundred feet long. And I’ve heard it described as kind of like an inside-out fire hose. Very strong and waterproof, but to some degree flexible.”

class="">The mess is another problem. The dust from the fresh cut concrete mixes with lubricating water to form a slurry that runs out of the slot. Concrete slurry isn’t good for the environment. It mucks up the water and changes the chemistry. So the slurry generated by the cutting process has to be captured and pumped to holding tanks. After the concrete particles have settled out, the water can be recirculated to control the dust and lubricate the wire as it cuts. And this whole process happens essentially non-stop. Time is of the essence so that the internal stress doesn’t close the slot while the wire is still inside it. Slot cutting is relatively low impact on the dam operations, but parts of the dam have to be shut down to avoid an accident like a broken wire being pulled into a hydro unit or spillway gate.

class="">One of the reasons this is possible at all is that TVA’s concrete dams experiencing ASR are all gravity dams. In essence, that means that any vertical slice of the dam is theoretically stable on its own without lateral support. Cutting a slot in an arch dam wouldn’t work, because they depend on axial thrust forces for stability.

class="">Before, during, and after the slot cutting operation, there’s an intensive monitoring program to keep an eye on how the dam is behaving and methodically measure the movement and strains to make sure the dam responds in the way the engineers predict.

class="">Kristen: “We have hundreds and hundreds of instruments on the concrete portion of the dam. We measure the slot that we’ve cut. Is it closing? Is it opening? At what rate is it closing or opening? We measure our spillway piers. Are they moving? We measure expansion joints. Everything in every direction we measure.”

class="">And those measurements are important because the slot cutting isn’t a one-time permanent solution. This doesn’t slow down the alkali-silica reaction in the concrete at all. It just mitigates the stress building up in the structure as the concrete expands, which is basically a non-stop process. Over time, the slots close. That means that TVA has to go through the operation regularly.

class="">Kristen: “Every approximately five years, we update, we use finite element analysis models on our concrete

growth projects. So they take all of those years of new information data from the instruments and they recalibrate and they rerun these models and they can tell us how effective the slot cut is. They can tell us when we need to do it again. Whatever we need to do to ensure that we are maintaining the integrity of our damms and the adjacent equipment, that's what we do."

class="">I was curious why they don't just cut a big slot to get a longer period of relief before having to do it again. In hindsight, it was kind of a dumb question:

class="">Kristen: "The simple answer is so we don't leave a big hole in the dam. The slot cut at Chickamauga is approximately a half and inch. It's a lot easier to stop water from flowing through a half an inch slot in a dam than it would be maybe a six inch wide slot. In addition, slot cutting is expensive."

class="">In other words, TVA wants to disturb their structures as little as possible, while still mitigating the problems AAR causes. It's a back-and-forth thing. You cut, observe, wait, and only cut again when it's necessary. It's good stewardship of the resources available to take care of the structures we've already built.

class="">Alkali-silica reaction in concrete is a huge problem. It's something engineers have to consider when designing basically any concrete structure, which means it's something that quarries, batch plants, testing labs, and contractors have to think about as well. Since the 1970s, we've gotten pretty good at avoiding it in our structures. But since it's often a slow-growing issue, we're still figuring out how to deal with the problems it's causing on the stuff we built before we really had a handle on it. On mass concrete structures, like TVA's dams, it could have been a death blow, significantly shortening the lifespans of these massive projects. But they figured out a creative solution to live with it.

class="">Kristen: "I mean, it's cool. And when you think about a dam, it's a water barrier. It is designed to hold back water. So the last thing you expect to do is to cut a piece out of it. But we do. We do."

(Edited for length)

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FEATURES

Raouf Chebri, Neon Tech Blog: Today, we are releasing Neon CLI, a command-line interface that enables developers to manage Neon resources directly from the terminal! With Neon CLI, you can handle authentication, and manage projects, branches, databases, roles, and much more without leaving your command line. Get started by installing the CLI using the following command: `Use neonctl auth command [...]`

Yan Cui, Neon Tech Blog: “Ephemeral environments” refers to the ability to create short-lived copies of your system so that: It’s a powerful practice and works great with services that charge on a pay-per-use basis. There is no extra charge for these environments – it’s only the activities that count. However, services that charge by uptime (aka “serverful”), such as [...]

Charmmie Hendon, Serverless Blog: I’m Charmmie Hendon, and I’m passionate about diversity and inclusion in tech. Here’s a bit of my personal experience, and how you too can affect change.

Arсени Kravchenko, Neon Tech Blog: Every now and then, people ask me: “I am new to agentic development, I’m building something, but I feel like I’m missing some tribal knowledge. Help me catch up!” I’m tempted to suggest some serious stuff like multiweek courses (e.g. by HuggingFace or Berkeley), but not everyone is interested in that level of diving. So [...]

Andrew Tate, Neon Tech Blog: “Timing is everything.” Some say that’s a quote from Shakespeare, but imho, the true bards of time are the 52 contributors to the `pg_cron` extension. The concept being `pg_cron` is simple: run scheduled jobs directly within Postgres using familiar cron syntax. The simplicity belies the power, though. With `pg_cron`, you can schedule database maintenance, automate [...]

Alex DeBrie, Serverless Blog: How to use the Serverless Event Gateway (part of the Serverless Platform) to build REST APIs and react to custom events.

Paul Scanlon, Neon Tech Blog: In this post, I’d like to introduce Neon’s serverless driver which is suitable for use with Vercel Edge Functions. I’ll explain how to use it with Next.js and a free Neon serverless Postgres database Here’s a demo I created. I’ve called it Ping Thing. What does Ping Thing do? When users click Ping, the app [...]

Stefanie Monge, Serverless Blog: How Serverless Partner AbstractAI leveraged the Serverless Framework and Lambda to reduce the cost of running back-end services by 95%.

Gareth McCumskey, Serverless Blog: Container Image Support for AWS Lambda has now been added. Read up on why you may (or may not) want to use it and how easy it is to use with the framework

Heikki Linnakangas, Neon Tech Blog: The idea behind Neon is to create a new serverless Postgres service with a modern cloud-native architecture. When building for the cloud it usually is a good idea to separate storage and compute. For operational databases such design was first introduced by AWS Aurora 1, followed by many others 2 3, however none of the implementations were [...]

Carlota Soto, Neon Tech Blog: Reddit’s answer for solving noisy neighbors is simple: Have a talk with them, if you can stay calm, be as straight forward as possible. Don’t bend on your position. At least find out what the other neighbors think. Good advice, but it can be difficult to talk to CPUs; they tend not to like it [...]

Carlota Soto, Neon Tech Blog: Database branching makes it easy to create database copies for your non-prod environments, but some companies face the obstacle of how to anonymize PII to ensure privacy. This can be solved with Neosync, an open-source tool that allows developers to anonymize data across branches. Try it with Neon. Developers worldwide are adopting Postgres to run [...]

Taraneh Dohmer, Neon Tech Blog: Neptune is an AI-powered concierge that helps couples navigate high-stakes financial decisions, starting with prenups. Neptune guides couples through an emotionally intelligent process that replaces the guesswork and complexity of prenups (because most of us are not lawyers) and helps couples identify their goals and priorities for their prenuptial agreement. Once couples have discovered their [...]

Mahmoud Abdelwahab, Neon Tech Blog: A few months back, we had an internal AI hackathon at Neon where we could learn and experiment with OpenAI’s APIs. One of the ideas was to build an online AI SQL playground for Postgres. It would enable developers to connect to a Neon Postgres database and have an AI assistant generate SQL queries using [...]

Evan Shortiss, Neon Tech Blog: Keycloak is an open-source identity and access management solution that centralizes authentication and authorization management. It provides features such as single sign-on (SSO), two-factor authentication, social login, and user federation with LDAP or Active Directory user federation. Various identity protocols such as Open ID Connect, SAML 2.0, and OAuth 2.0 are supported, which makes integrating [...]

Pedro Figueiredo, Neon Tech Blog: We just launched Neon Auth and of course the immediate follow-up task I had was “let’s teach my AI tools to use it for me”. So we’ve been wiring Neon Auth into our MCP so you can open Cursor, Claude, or your IDE or choice and let the AI build a backend for you. Here’s [...]

Carlota Soto, Neon Tech Blog: Neon offers the most powerful Postgres API on the market for applications that need to provision and manage thousands of databases programmatically. Whether you’re building a platform (like Retool) or deploying infrastructure through AI agents (like Replit), Neon gives you the tools to do it efficiently and at scale. Why is that? It starts with [...]

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